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Mineral Precipitation and Geometry Alteration in Porous Structures: How to Upscale Variations in Permeability−Porosity Relationship?

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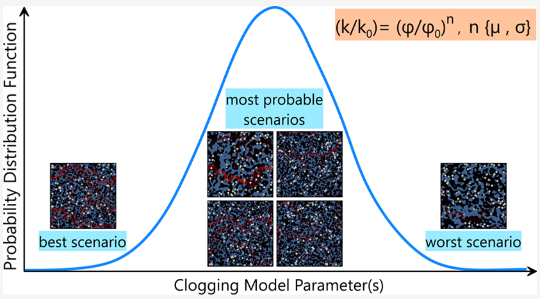
ABSTRACT: Porous materials in natural and engineered environments are subject to morphological changes resulting from interacting chemical and physical processes. The intricate nature of these coupled processes, occurring at various temporal and spatial scales, poses challenges in predicting alterations in porosity and permeability. Delineating the controls of mineral precipitation reactions is particularly challenging because it requires the implementation of nucleation criteria and growth mechanisms. By conducting pore-scale simulations, we investigated the impact of the amount and stochastic distribution of crystallites, controlled by nucleation, on pore geometry and permeability in two-dimensional porous structures. The observed relationships between porosity and permeability exhibit characteristics that differ from the ones that are typically applied in dissolving porous media because of the clogging effect. Additionally, we propose a stochastic framework that upscales the coevolution of permeability and porosity across length scales. This framework enables the upscaling of clogging behavior to continuum-scale simulations based on statistical probability distributions of permeability−porosity variations.



1. INTRODUCTION

Many natural and engineered systems and coupled processes have been typically modeled by reactive transport models (RTM) at the continuum scale. With the continuum assumption, averaged properties are assigned in each grid cell. The averaged values are often estimated, i.e., upscaled from the smaller scale features. Upscaling can be achieved by mathematically deriving macroscale equations from microscale principles,[1] or by developing constitutive relations from finer scale simulations and experiments.

Porosity and permeability are two of the most important and common continuum scale parameters in hydrological and hydrogeochemical modeling. Porosity provides a homogenized representation of the pore structures, and permeability is usually upscaled based on permeability−porosity relations. These relations do not explicitly capture the pore-scale complexities and microscale features and only provide approximations at the continuum scale. Establishing a relationship between porosity and permeability in porous media to preserve as much as pore-scale dynamics is an increasingly important area of research, particularly with recent advancements in computation and imaging technologies. Research to establish a correlation between porosity and permeability can be broadly categorized into two groups. The first category relates the permeability of an intact medium to its porosity, commonly found in the field of digital rock physics.[2−6] The second category focuses on linking changes in permeability to alterations in porosity, i.e., the investigation of permeability evolution[7−10] that can be caused by a range of coupled thermal, hydraulic, mechanical, and chemical (THMC) processes. Despite their similar goals, these two categories are fundamentally different. Still, porosity-permeability relations developed from these two types of studies are mostly used interchangeably. This paper focuses on the changes in permeability and porosity and their relationship in porous media.

The morphology changes of porous materials (e.g., fractured geomaterials, fuel cells, and brain tissues) are influenced by interacting chemical−physical processes such as dissolution, precipitation, biomass accumulation, swelling, shrinking, and mechanical compaction. The intimate interplay between different factors affects permeability changes. For instance, during dissolution and precipitation, different reactive regimes characterized by Peclet (Pe) and Damkohler (Da) numbers affect the relationships between porosity and permeability.[11−13] In turn, anisotropic rock texture and preferential flow paths can also significantly impact dissolution−precipitation patterns.[14−17]

Specifically, the precipitation of secondary phases during reactive transport is a complex phenomenon in various geological and environmental settings. Substances such as

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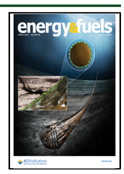

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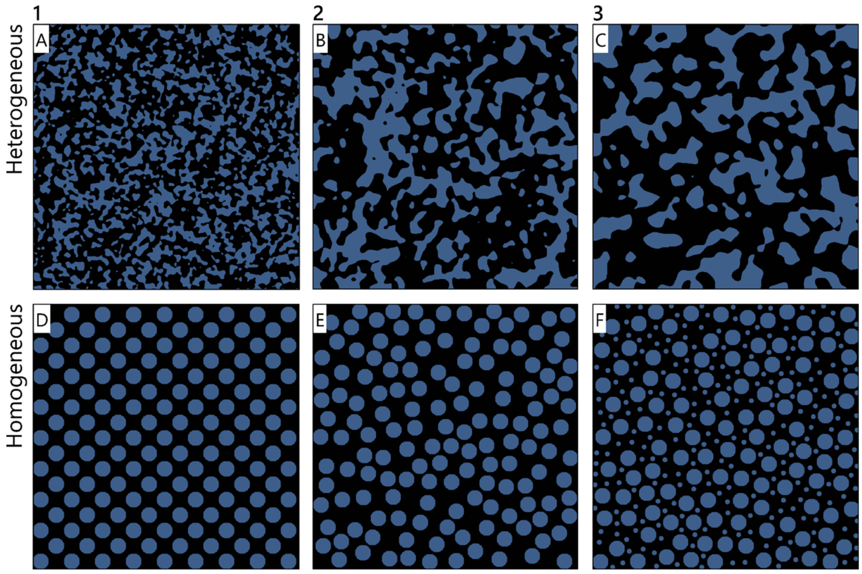

Figure 1. Selection of random two-dimensional porous geometries, representing natural rock geometries (A−C, heterogeneous) and simplified ordered/disordered porous structures (D−F, homogeneous). The darker (black) and lighter (blue) colors represent pore space and grains (solid phases).

hydrate,[18] asphaltene,[19,20] biomass,[7] and minerals[21−24] can deposit in porous media and clog pore spaces differently. Among these substances, minerals have an added level of complexity because of nucleation, the pregrowth phenomenon that determines the starting time and location of mineral growth. Studying mineral precipitation in natural and engineered porous materials has broad implications for addressing several United Nations' Sustainable Development Goals, including clean water and sanitation, climate action, and affordable and clean energy. Mineral precipitation can significantly influence the behavior of contaminants in the environment by altering their solubility and mobility, thereby affecting their transport and fate. The sequestration of inorganic pollutants (such as arsenic, antimony, selenium, and other heavy and toxic metals) in carbonates[25−29] and retention of radium by its coprecipitation with barium sulfate to treat contaminated water, water waste, and industrial waste[30−32] are only a few examples. Mineral precipitation in porous rocks plays a vital role in addressing several geo-energy challenges,[33] such as scale formation in geothermal systems,[34] injectivity impairment due to salt aggregates during CO_2 storage,[24,35,36] and clogging of reservoir rocks during enhanced hydrocarbon recovery. Favorable effects of mineral precipitation include CO_2 mineral trapping in basalt[37] and fracture healing of caprock leakage pathways in storage sites.[23]

Essentially, mineral nucleation is a critical factor in controlling the distribution of minerals over time and space.[8,22,38−43] The interplay between solute transport, nucleation, growth, and porous media structure is intricate. Several experimental and numerical studies have shown that the same amount of porosity reduction due to mineral precipitation might result in different permeability reductions.[8,14,49−51,16,22,39,44−48] Furthermore, arbitrary selection of clogging models in RTMs may lead to considerable errors in predicted permeability for the same amount of porosity loss.[7,52] Despite all the mentioned variations and uncertainties, the permeability−porosity evolution during RTM in the continuum scale is usually oversimplified and handled with a single equation with a constant variable. Therefore, a process-based approach is necessary when analyzing permeability−porosity evolution.

In this work, we study the effect of crystallite distribution (controlled by nucleation) on pore geometry alteration and, consequently, permeability evolution in diverse two-dimensional porous media. In addition, for the first time, we propose a framework for upscaling the coevolution of permeability−porosity, which combines simulations at both the pore and continuum scales to produce a more accurate representation of permeability changes in porous media. This study enables a comprehensive delineation of the clogging behavior in porous media, which can inform decision-making in various applications, such as CO_2 storage, water treatment, and geo-energy resources.

2. MATERIALS AND METHODS

2.1. Selection of Porous Geometries. Six different two-dimensional (2D) porous geometries were generated using PoreSpy[53] (Figure 1) to perform pore-scale reactive transport modeling (RTM). The synthetic porous media structures encompass (a) highly heterogeneous structures that mimic natural rock geometries with different permeability and anisotropies (Figure 1, top row) and (b) simplified homogeneous geometries with ordered and disordered arrangement of spheres (Figure 1, bottom row). The pore volume (porosity) for all cases is chosen to be approximately 60% (59.6−60.7%) because the percolation threshold of a 2D square lattice with site percolation is found to be 59.27%.[54] The initial permeability of the structures varies from 1.57×10^{-10} to 2.59×10^{-9} m^2. In the text, we refer to each geometry in Figure 1 as Het (heterogeneous, top row) or Hom (homogeneous, bottom row), followed by their number. For instance, Het_2 for heterogeneous medium number 2. The properties of the structures are summarized in Table S1.

2.2. Model and Assumptions. We used the lattice Boltzmann method (LBM) for RTM and flow simulations with the D2Q9 lattice scheme. The advection−diffusion reaction equation was used for tracking the concentration of different species:

$$\frac{\partial C_j}{\partial t} + \nabla \times (-D_j \nabla C_j + uC_j) = R_j \quad (1)$$

where C_j (NL^{-3}) is the aqueous concentration of species j, D_j (L^{2}T^{-1}) is the diffusion coefficient of species j in water, u is velocity (LT^{-1}), and R_j (NL^{-3}T^{-1}) is the source or sink term due to reactions for species j. The

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discretized LB equation used to solve the ADR equation (the mass transport) is as follows:

$$g_i^j(x + c_i\Delta t, t + \Delta t) = g_i^j(x, t) - \frac{\Delta t}{\tau_g}[g_i^j(x, t) - g_i^{eq,j}(x, t)] + \Delta t Q_i^j(x, t) \quad (2)$$

$$g_i^{eq,j} = w_i C^j\left(1 + \frac{c_i \times u}{c_s^2}\right) \quad (3)$$

where g_i^j is the discrete distribution function, $g_i^{eq,j}$ is the equilibrium distribution function, and C^j is the concentration of species j. Δt is time resolutions, τ_g is relaxation time, c_s is lattice speed of sound, and c_i and w_i are discrete velocity sets and weighting coefficients. For eq 2 to correctly recover the ADR equation as defined by eq 1, the Chapman−Enskog analysis dictates that the following relation must exist between the diffusion coefficient and the relaxation parameter:

$$D = c_s^2\left(\tau_g - \frac{\Delta t}{2}\right) \quad (4)$$

The D2Q9 lattice scheme used, where 2 is the number of spatial dimensions (x and y) and 9 is the number of discrete velocities, $w_0 = 4/9$, $w_{1-4} = 1/9$, and $w_{5-8} = 1/36$ and

$$c_i = \begin{cases} (0, 0), \ i = 0 \\ \left(\cos\theta_i, \sin\theta_i\right), \ \theta_i = \left(i - 9\right)\frac{\pi}{2}, \ i = 1 - 4 \\ \left(\cos\theta_i, \sin\theta_i\right)\sqrt{2}, \ \theta_i = \left(2i - 9\right)\frac{\pi}{4}, \ i = 5 - 8 \end{cases} \quad (5)$$

After solving eq 2, the concentration of species j is calculated as follows:

$$C^j = \sum_{i=0}^{8} g_i^j \quad (6)$$

The source/sink term is given by

$$Q_i^j(x, t) = w_i[q_N^j(x, t) + q_G^j(x, t) + q_S^j(x, t)] \quad (7)$$

$$q_N^j = \frac{dC_N^j}{\Delta t} \quad (8)$$

$$q_G^j = \frac{dC_G^j}{\Delta t} \quad (9)$$

$$q_S^j = D_j(SR^0 - SR), \ \text{if } SR < SR^0 \quad (10)$$

The subscripts N, G, and S in eq 7 denote the nucleation, growth, and infinite source of the solution on top of the substrate. The simple reaction of $\{A_{(aq)} \rightleftarrows A_{(s)}\}$ is considered:

$$R_G = k_G S(1 - SR) \quad (11)$$

where R_G ($mol \cdot s^{-1}$) is the growth or reaction rate, k_G ($mol \cdot m^{-2} \cdot s^{-1}$) is the growth rate constant, S (m^2) is the reactive surface area, and SR is the saturation ratio:

$$SR = \frac{C_A}{C_{eq}} \quad (12)$$

where C_A [$mol \cdot L^{-1}$] is the concentration of mineral A in the aqueous phase, and C_{eq} [$mol \cdot L^{-1}$] is the equilibrium aqueous concentration of mineral A. The reactive surface area, S, is updated at each time step, assuming that a crystal grows in the middle of a grid as a cube, as depicted in Figure S1 (Supporting Information). The molar amount of precipitated material is used to calculate the reactive surface area, considering the molar volume of the growing crystal.

$$\delta x_c = \sqrt[3]{n_c \times V_m} \quad (13)$$

where δx_c is the edge length of a cubic crystal, n_c is the molar amount of the precipitated crystal within the grid, and V_m is the molar volume of the precipitated phase. This approach acknowledges the presence of a substrate to which one side of the crystal is attached, as depicted in Figure S1

$$S = 5(\delta x_c)^2 + S_{rn} \quad (14)$$

In eq 14, the subscript rn represents *reactive neighbors*. When a grid is fully precipitated, it can provide reactive surface area for its neighboring grids. Therefore, S_{rn} denotes the reactive surface areas contributed by the neighboring grids that are fully precipitated. If, for instance, ΔX represents the grid width and there are two neighboring grids that are already fully precipitated, then $S_{rn} = 2\Delta X^2$. In the absence of fully precipitated neighboring grids, $S_{rn} = 0$. Indeed, as the crystals continue to grow, their surface area increases, leading to a larger reactive surface area, which enhances the rate of reactions.

To communicate between the reaction and the transport processes, we employ an operator-splitting method in a sequential, noniterative manner. After completing each transport step, the code initiates the reaction sequence to determine the source/sink terms. Initially, concentrations are transformed from LB units to physical units using a conversion factor. We then calculate the concentration differences in physical units. As a final step, we convert these differences back into LB units, acquiring the concentration differences as required in eqs 8 and 9. A comprehensive discussion of the unit scales and conversion factors s presented in Chapter 7 of Krüger et al.'s book.[55]

The model has been validated by comparing its results with the analytical solution of a one-dimensional diffusion-controlled growth of a planar solid with Stefan moving boundary condition[56] (Figure S2). Additionally, to prevent operator splitting errors and concentration overshooting, we have ensured that the Courant and von Neumann criteria are satisfied. Further information and results can be found in the Supporting Information section.

The simulation domain was a quasi-2D system of 500 × 500 × 1 grid cells with a spatial resolution of 20 μm. The periodic boundary condition was applied for all the boundaries. The domain was initially water-saturated. There is no advection in the system, and an infinite source of the oversaturated solution was imposed on top of the domain via a source term in the discretized lattice Boltzmann (LB) equation. More details of the LBM model, mineral growth (reaction) model, and subgrid assumptions and implementations can be found in our previous studies.[8,39,42,43,57]

This work did not use the entire nucleation model criteria. Instead, a predefined number of crystallites (as a proxy for stable nucleation sites) were populated randomly for initializing the system, as the main focus is tracking and quantifying the consequences of mineral precipitation on the hydrodynamics of porous media. Each pore lattice was assumed to be on a homogeneous nonreactive solid substrate surface, as schematically depicted in Figure S1. The crystallites may form everywhere within the entire pore space, but each grid cell can only have one crystallite. This is equivalent of only considering heterogeneous nucleation events, which is nucleation on specific nucleation sites, surfaces, or substrates. The reaction rate afterward controlled the continued growth of the introduced crystallites until reaching the clogging state.

To encompass a comprehensive range of scenarios, varying by orders of magnitude from sparse to densely populated systems, we chose to numerically model three distinct crystallization cases. These cases, involving 10, 100, and 1000 crystallites respectively, were applied to each of the six porous media geometries examined in our study. For instance, the case with 1000 crystallites can replicate conditions akin to salt aggregation near well areas during CO_2 storage, where the frequent interaction among crystallites can lead to rapid and intricate clogging behaviors. Conversely, the scenario with 10 crystallites can be representative of a system with limited nucleation sites and minimal interactions, analogous to slower carbonate precipitation processes. To ensure robustness, each case was simulated with eight stochastic realizations. This yielded a comprehensive data set comprising a total of

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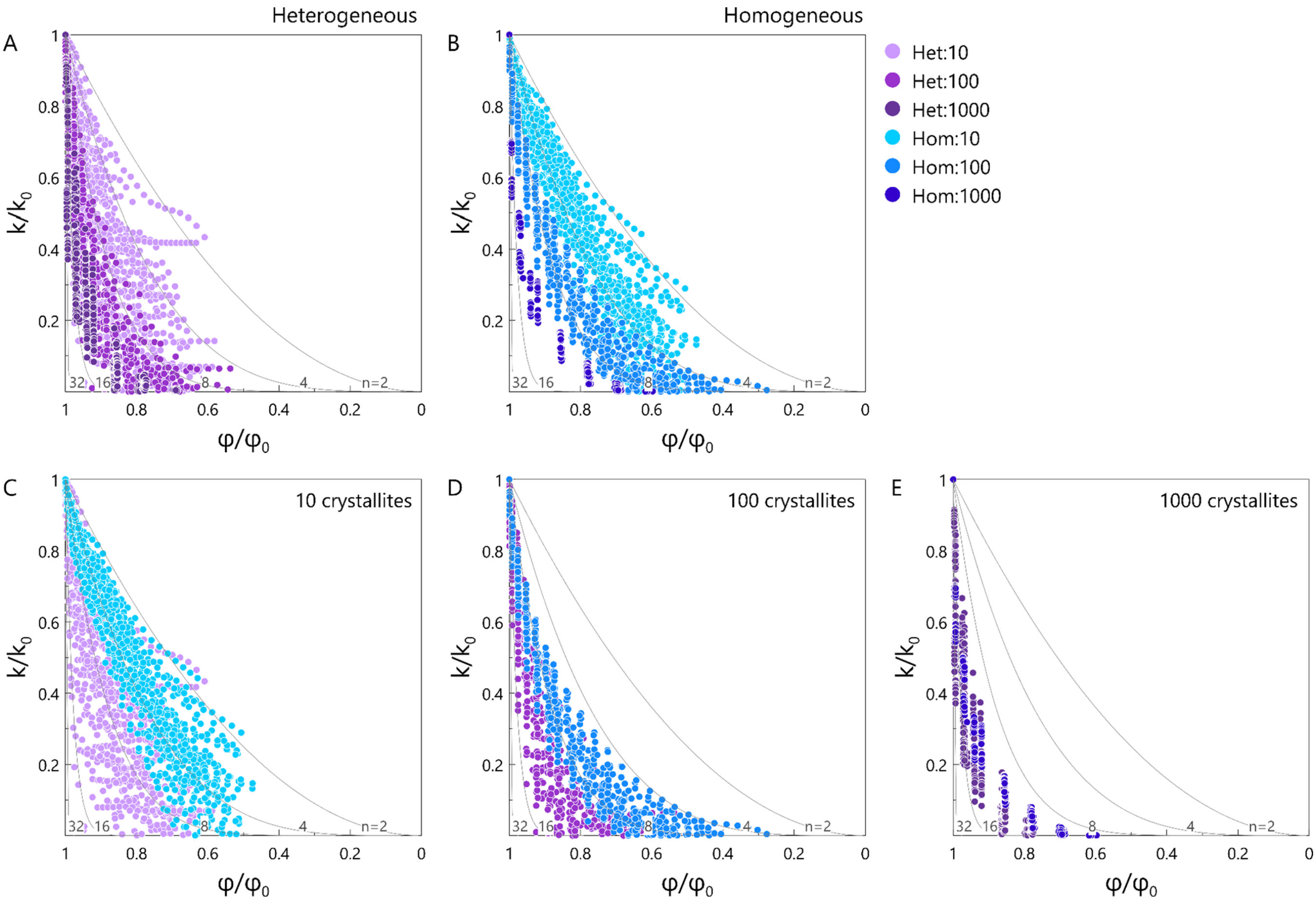

Figure 2. Variations in flow properties (expressed as normalized permeability−porosity relationship) of evolving two-dimensional porous geometries induced by mineral precipitation. Two structural classes, six different porous geometries, and three crystallization cases (10, 100, and 1000) are shown. Power-law bounds with different exponents ($n = 2^x$ [$x = 1,...,5$]) are also plotted. (A) Heterogeneous and (B) homogeneous and porous geometries. Variations of decline in normalized permeability−porosity during stochastic growth of (C) 10, (D) 100, and (E) 1000 crystallites. Violet: heterogeneous; blue: homogeneous cases.

144 simulations (6 geometries × 3 crystallization cases × 8 repetitions). In our discussions, we uniquely label each case with the respective crystallite count, followed by 'c'; for example, '100c' signifies the scenario involving 100 crystallites.

Crystallites have the same size (1×10^{-9} m radius). Since the primary objective of this work is to investigate the impact of crystal distribution on changes in permeability, a similar growth rate constant was considered for all the cases (1.5488×10^{-6} mol·m^{-2}·s^{-1}). To prevent the introduction of unrealistic or arbitrary input values, we opted for a specific rate constant of calcite precipitation, which is derived from the neutral mechanism and documented by Palandri and Kharaka[56] as $\mathrm{Log}(k) = -5.81$. To examine the impact of reaction rate on the outcomes, we conducted additional simulations with $k = 1.5488 \times 10^{-5}$ mol·m^{-2}·s^{-1}, an order of magnitude higher than the original value, for Het_1 and Hom_1 geometries.

The permeability of the evolving domains was calculated using LBM for hydrodynamics. We used Yantra open source code[58] to calculate the velocity fields at the x and y directions. Periodic and no flow boundary conditions were applied to the boundaries along and perpendicular to the flow direction, respectively. To recover the hydrodynamic properties in single-phase fluid flow, continuity equation (eq 15) and Navier−Stokes equation (eq 16) should be solved simultaneously. To do so, the LB model in eq 17 was used.

$$\nabla \times u = 0 \tag{15}$$

$$\rho\frac{\partial u}{\partial t} + \rho u \times \nabla u = -\nabla P + \mu\nabla^2 u + F \tag{16}$$

where ρ (ML^{-3}) is density, u (LT^{-1}) is velocity, P (M^{1}L^{-1}T^{-2}) is pressure, μ (M^{1}L^{-1}T^{-1}) is dynamic viscosity, and F (M^{1}L^{-2}T^{-2}) is body force.

$$f_i(x + c_i\Delta t,\ t + \Delta t) = f_i(x,\ t) - \frac{\Delta t}{\tau}(f_i(x,\ t) - f_i^{\mathrm{eq}}(x,\ t)) + \left(1 - \frac{\Delta t}{2\tau}\right)F_i(x,\ t) \tag{17}$$

$$f_i^{\mathrm{eq}} = w_i\rho\left(1 + \frac{u\cdot c_i}{c_s^2} + \frac{(u\cdot c_i)^2}{2c_s^4} - \frac{u\cdot u}{2c_s^2}\right) \tag{18}$$

$$F_i = w_i\left(\frac{c_i\cdot F_\alpha}{c_s^2}\right) \tag{19}$$

$$c_s^2 = \frac{1}{3}\left(\frac{\Delta x}{\Delta t}\right)^2 \tag{20}$$

where Δt and Δx are time and space resolutions. f_i is the discrete distribution function, f_i^{eq} is the equilibrium distribution function for hydrodynamics. F_i is body force, c_s is lattice speed of sound, u is velocity, ρ is density, and τ is relaxation time for hydrodynamics. It should be noted that the lattice speed of sound, c_s, can vary based on the chosen lattice structure. Because we used the same lattice structure (D2Q9), weighting coefficients (w_i), and discrete velocity sets (c_i) for ADR LB and hydrodynamics LB, the lattice speed of sound in eq 20 can be used

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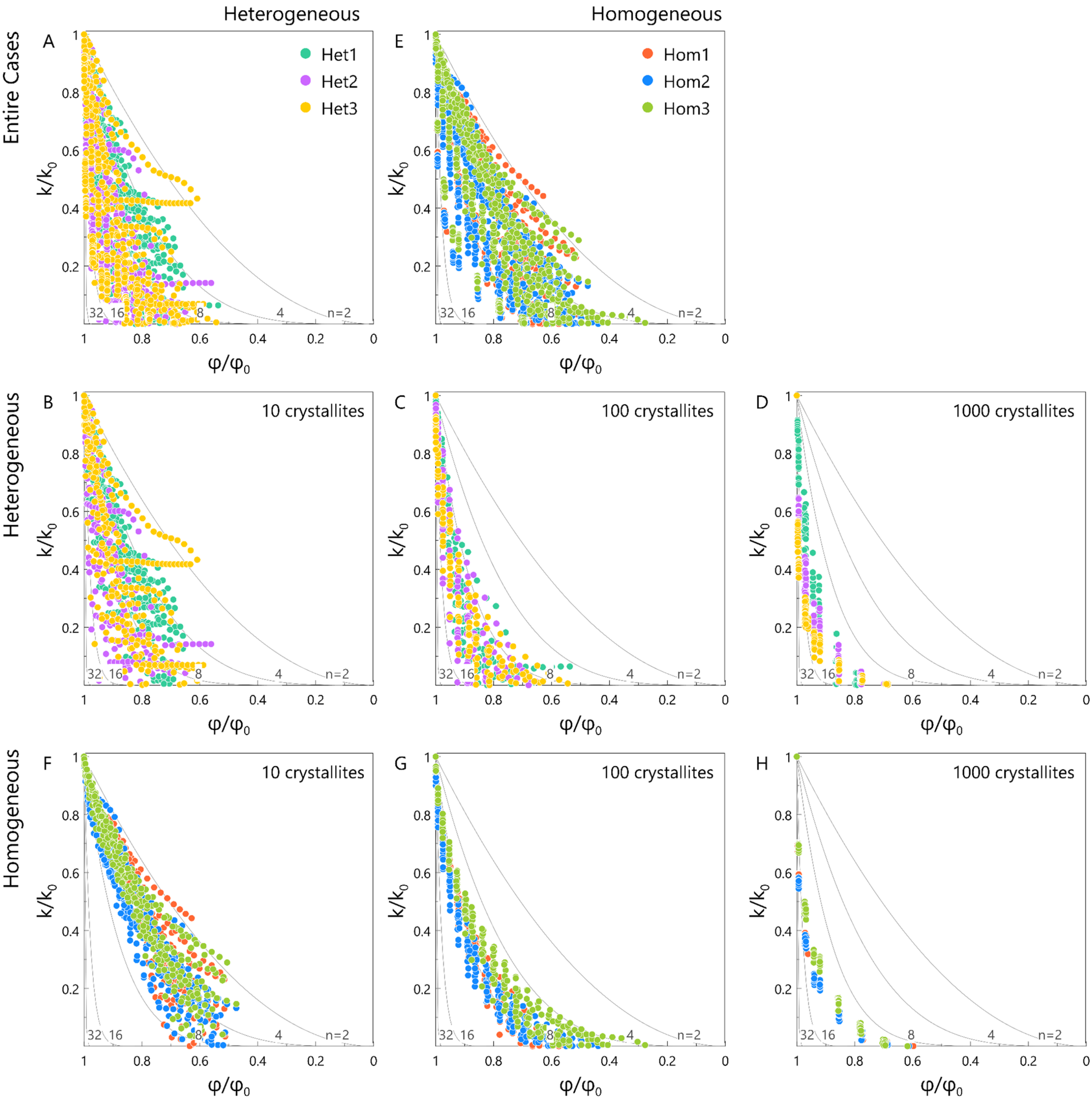

Figure 3. Variations in flow properties (expressed as normalized permeability−porosity relationship) of evolving two-dimensional porous geometries induced by mineral precipitation. Two structural classes, six different porous geometries, and three crystallization cases (10, 100, and 1000) are shown. Power-law bounds with different exponents ($n = 2^x$ [$x = 1,...,5$]) are also plotted. Variations in heterogeneous geometries for (A) entire crystallization cases, (B) 10, (C) 100, and (D) 1000 crystallites. Variations in homogeneous geometries for (E) entire crystallization cases, (F) 10, (G) 100, and (H) 1000 crystallites.

for both eq 3 and eq 18. After solving eq 16, density and velocity can be calculated from the following equations:

$$\rho = \sum_{i=0}^{8} f_i^j \tag{21}$$

$$u = \frac{1}{\rho}\sum_i f_i c_i + \frac{F_\alpha}{2\rho} \tag{22}$$

Darcy's law was used to calculate the permeability of the porous media:

$$\bar{J} = \frac{-k}{\mu}\nabla P \tag{23}$$

where $\bar{J}$ (LT^{-1}) is average flux (flow rate per unit area), k (L^2) is permeability, μ ($M^1L^{-1}T^{-1}$) is dynamic viscosity, and ∇P ($M^1L^{-1}T^{-2}$) is pressure gradient. ∇P in eq 23 can be replaced by F, the body force, in eq 16.

3. RESULTS AND DISCUSSION

Figure 2 compares the normalized permeability−porosity relationships (k/k_0 plotted against ϕ/ϕ_0) from six different

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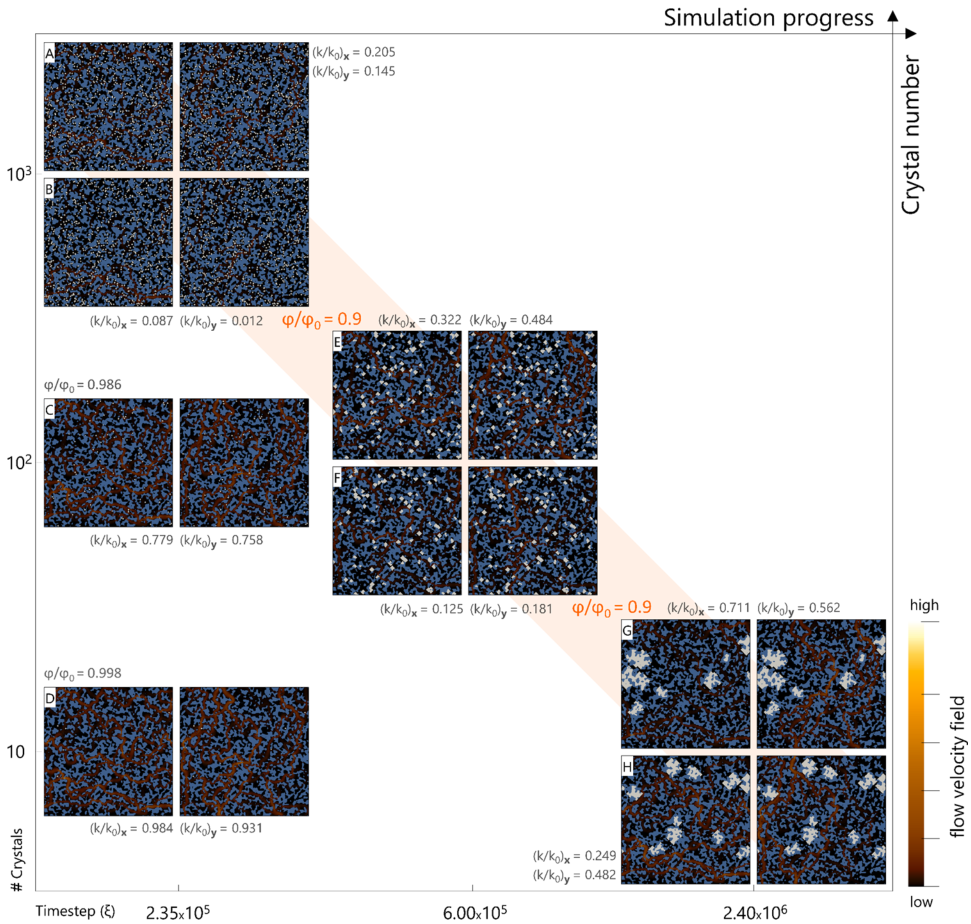

Figure 4. Pore morphology alterations because of mineral precipitation and growth for pseudorock geometries (heterogeneous 1 media in Figure 1). Besides the 2D surface growth stochastic realizations, normalized porosity loss (ϕ/ϕ_0), normalized permeability decline (k/k_0) in x,y-directions, and flow velocity fields are indicated as a function of crystallites number and simulation progress. For each part, the velocity map to the left and right represent the x and the y directions, respectively. The geometries on the diagonal have similar normalized porosity loss (ϕ/ϕ_0 = 0.9). (A) 1000 crystals—minimum permeability decline. (B) 1000 crystals—maximum permeability decline. (C,D) 100 and 10 crystals, respectively, at the same time step as A and B. (E) Continuation of C—100 crystals, minimum permeability decline. (F) 100 crystals—maximum permeability decline. (G) Continuation of D—10 crystals, minimum permeability decline. (H) 10 crystals—maximum permeability decline.

porous geometries, and three crystallization cases (10, 100, and 1000 crystallites) with power-law formulations with different exponents (k/k_0 = $(\varphi/\varphi_0)^n$). Overall, pore morphological alteration from mineral precipitation and growth affected heterogeneous geometries (representing real rock structures) more substantially than homogeneous structures. The difference between different numbers of precipitated crystallites is more distinct for homogeneous systems, even at early times (Figure 2). The entire simulation results for each case are shown in Figures S3 and S4, where each curve represents one single simulation with the initial position of crystallites varied randomly. Changes in permeability because of mineral precipitation and porous geometry alteration are investigated as a function of (a) total porosity changes (expressed as φ/φ_0) in Figures 2, 3 and S3, and (b) reaction progress (ξ) in Figure S4.

Permeability impairment intensifies with a higher count of crystals. This can be observed from the steeper slopes of each simulation case in Figures S3 and S4. Furthermore, upon closer examination of Figures 2 and 3, a notable observation emerges: the data points for the 10c case fall within the fitting lines of power-law functions with n = 2−16, while those for the 10^2c case are confined to n = 4−16, and for the 10^3c case, they span from n = 10−16. The distinction between heterogeneous and homogeneous geometries is particularly pronounced in 10c and 10^2c cases, with the data points falling within curves with smaller n values in the homogeneous cases. In the 10^3c cases, the difference between the heterogeneous and homogeneous cases is less significant.

The scattering in the data points is largely caused by the stochastic realizations of the 10c and 100c cases compared to the 1000c cases (Figure 2). This observation is also corroborated by the covariance values between the porosity and permeability, which are 0.042, 0.035, and 0.03 for the 10c, 100c, and 1000c cases, respectively.

Figure 3A−D and E−H present a comparative analysis of results for heterogeneous and homogeneous systems, respectively, to focus on the impacts of initial geometries on permeability evolution. Figure 3A−D shows that the rate of permeability reduction, especially at the early stage, increases from Het_1 to Het_3, as also confirmed in Figure S3. It might seem unexpected since Het_3 has more prominent flow pathways and higher initial permeability, being 1 order of magnitude higher than that of Het_1. Precipitation-included morphology alteration caused greater permeability impairment in Het_3 given the same number of initial crystals. Het_3 features massive but fewer highways (compared to others),

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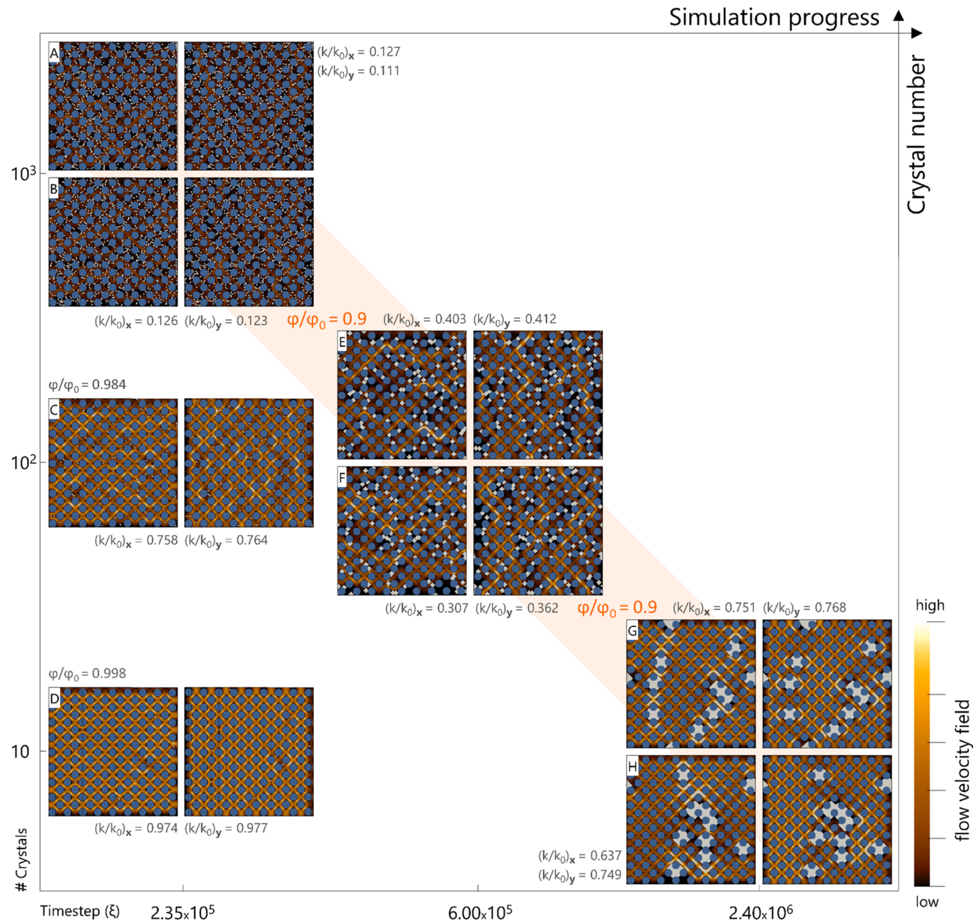

Figure 5. Pore morphology alterations as a result of mineral precipitation and growth for simplified geometries (homogeneous 1 media in Figure 1). Besides the 2D surface growth stochastic realizations, normalized porosity loss (ϕ/ϕ_0), normalized permeability decline (k/k_0) in x,y-directions, and flow velocity fields are indicated as a function of crystallites number and simulation progress. For each part, the velocity map to the left and right represent the x and the y directions, respectively. The geometries on the diagonal have similar normalized porosity loss (ϕ/ϕ_0 = 0.9). (A) 1000 crystals—minimum permeability decline. (B) 1000 crystals—maximum permeability decline. (C,D) 100 and 10 crystals, respectively, at the same time step as A and B. (E) Continuation of C—100 crystals, minimum permeability decline. (F) 100 crystals—maximum permeability decline. (G) Continuation of D—10 crystals, minimum permeability decline. (H) 10 crystals—maximum permeability decline.

meaning that small patches of solid precipitation can break these fluid pathways into several smaller and less transmissive channels. One might therefore conclude that heterogeneous media with few preferential flow paths are prone to more severe permeability impairment. For instance, fracture permeability can drop significantly and quickly, even with a tiny amount of mineral precipitation. This suggests that the initial state of the porous geometry can play a critical role in the extent of permeability impairment caused by mineral precipitation.

Although Hom_1 and Hom_2 geometries are different and represent ordered and disordered distributions of circular elements, their corresponding changes in permeability are approximately similar, especially in the 100c and 1000c cases (Figure 3E−H). It implies that studying geometry alteration in 2D microfluidic systems with similar ordered and disordered disks in Figure 1 may lead to similar outcomes regarding permeability evolution. Furthermore, the permeability reduction rate for Hom_3 geometry, which introduced more heterogeneity in the pore-size distribution but likely more flow paths, is less severe than in the other two homogeneous cases (Figure 3G,H).

It should be noted that the observed variations in permeability trendlines are also associated with the characteristics of 2D or pseduo2D systems (e.g., flow between sedimentary lamina or in fractures). For instance, in the heterogeneous geometries (Figure 1), a handful of pore-throat locations control the overall flow and transport. In other words, there are few flow highways (the most transmissive flow paths) within the system, blocking which results in significant permeability decline. As such, when the system is heterogeneous and anisotropic, some sharp feedback (permeability decline) can be expected with minor changes in the system (porosity alteration). When there are fewer crystallites, the differences between realizations are more significant, as more spatially distributed nucleation is likely to smooth out the variations in permeability evolution.

The severity of permeability decline and degree of variations is expected to be lower for three-dimensional porous geometries, particularly for a limited number of growing crystals. This lower impact is attributable to several factors inherent in three-dimensional systems: a greater number of potential fluid pathways allows for easier bypassing of clogged zones; the altered surface area-to-volume ratio in 3D environments provides expanded space for crystal nucleation and growth; and the lower percolation thresholds of 3D systems indicate a higher tolerance for clogging before experiencing a critical reduction in permeability.

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The results of the sensitivity analysis on reaction rate are reported in Figures S5 and S6. The results did not reveal significant differences compared to the simulations primarily discussed in the paper (Figures S3 and S4).

Figures 4 and 5 show flow velocity maps, mineral precipitation patterns, and directional permeability values (k_x and k_y), and thus provide more detailed information regarding the spatial (pore volume change) and temporal evolution in the heterogeneous (Het_1) and homogeneous (Hom_1) systems, respectively. The x-axis shows the number of steps, and the y-axis shows the number of crystallites. Blue represents the grains, and white represents the precipitants. Two different probabilistic realizations, giving maximum and minimum permeability decline, are shown to demonstrate the effect of the spatial distribution of crystals. For example, subfigures E and F are two realizations with different spatial distributions of 100 crystallites. Here, we focus our comparison between the realizations in two instances: (a) at $t = 2.35 \times 10^5$ steps (A, B, C, and D) and (b) at $\phi/\phi_0 = 0.9$ (diagonal).

It is shown that the increasing number of growing crystals significantly contributes to a faster decline in permeability over time (Figure 3). The precipitation maps and directional flow velocity profiles further illustrate how for the same porosity loss ($\varphi/\varphi_0 = 0.9$), a higher number of crystals can lead to a larger permeability decline. With a larger number of crystals, it is more likely to clog flow pathways earlier and thus with less precipitation. In contrast, fewer crystallites require that the crystallites grow to a sufficiently large size to block flow pathways. Consequently, with similar growth dynamics, the corresponding ξ for 10, 10^2, and 10^3 crystallites to reach the same porosity reduction are 2.40×10^6, 6.00×10^5, and 2.35×10^5, respectively.

The effect of crystal size and distribution on permeability evolution is also evident in the comparison at a given time (e.g., 2.35×10^5 steps) when individual crystals have similar sizes. In Figure 4D, 10 growing crystals occupied around 0.2% of the pore volume resulting in 1.5 and 6.9% permeability reduction in the x and y directions, respectively. As is illustrated in Figure 4, at a similar reaction progress (2.35×10^5 steps), porosity changes of $\phi/\phi_0 = 0.986$ and 0.9 were observed for the 100 and 1000 crystallite cases, respectively, causing an average of 24 and 80% permeability decline (99% in the worst-case scenario, Figure 4B).

The clustering and spread of initial crystals can also affect permeability in different directions. For example, Figure 4G,H represent two different distributions of 10 crystallites at the same step. For the case shown in Figure 4G, the permeability reduction ratio in the y direction is smaller than that in the x direction ($k/k_0 y = 0.48 > k/k_0 x = 0.248$). This is because the crystals to the left are clustered along the same highway in the y direction but spread out along different highways in the x direction, and growth that blocks several highways in the x direction only blocks a few in the y direction. In contrast, in the case depicted by Figure 4H, the crystals are spread along the same highways in the x direction, resulting in greater permeability reduction in the y direction. Similar behavior can be observed in homogeneous cases (see Figure 5H as an example), but the difference in permeability ratio values is less pronounced.

Overall, as the system evolves and porosity decreases, the difference between permeability values across realizations increases (Figures 2 and 3). For a higher number of crystals, the size of individual crystals is small at early times, so they cause limited and similar levels of permeability impairment. However, as the precipitates aggregate or become big enough to block a highway, their distribution can make more significant differences. Conversely, the mass is divided between fewer crystals for heterogeneous systems with fewer crystals. Therefore, based on their distribution, they can cause a considerable or minimal permeability reduction, and the standard deviation in permeability values is considerable from the beginning for these cases.

To provide readers with a clear idea of the changes in velocity field for all geometries, visualizations of crystal formation and velocity fields for each geometry have been presented in Figure S7. These visualizations depict the spatial profiles of 100 crystallites after 6e5 timesteps.

4. UPSCALING

The conventional approach to addressing the impact of precipitation reactions on flow and transport processes involves employing simplified permeability−porosity relationships. Empirical, experimental, or theoretical models such as the Kozeny−Carman,[59,60] Verma−Pruess,[61,62] and power law (Table S2) are commonly utilized due to their practicality and ease of use. Moreover, they are integrated into the majority of commercial or open-access simulators, catering to a broad spectrum of geo-energy and geo-environmental applications.

In the models outlined in Table S2 (refer to Supporting Information), the parameter "n"' serves as a tuning factor, either informed by empirical data, extrapolated from smaller-scale simulations, or obtained from literature sources. When conducting large-scale simulations, a medium-specific value of "n" is allocated to individual grid cells, drawing from the property map of the domain, such as rock types.

The power value "n" exhibits a broad spectrum of reported values[9] and can be correlated to factors such as mineralogy, clay mineral fractions, clay surface area, pore size distribution, heterogeneity of the pore structure, and transport regime. Opting for a fixed "n" value may not be suitable for accurately characterizing permeability−porosity changes.[9] The predictability of permeability−porosity relations becomes increasingly uncertain when dealing with natural rocks containing diverse minerals with varying kinetics and subjected to different flow and transport conditions. Furthermore, as previously mentioned, a multitude of processes can induce distinct alterations in pore morphology, which are influenced by both the initial pore structure and the specific physical and chemical conditions governing the process. Additionally, our pore-scale simulations demonstrate that even with identical porosity changes resulting from the same process and initial geometry, different alterations in permeability may occur in precipitation-dominated systems. In other words, the spatial distribution of the deposited solid significantly influences the hydrodynamics of porous media. Consequently, capturing the absolute evolution of permeability−porosity with a single clogging model is nearly impossible. Conversely, integrating excessive complexities into a clogging model proves impractical, inefficient, and arguably unnecessary, especially when confronted with systems comprising millions of grid cells.

To address these challenges, we propose a hierarchical upscaling approach tailored to accommodate the inherent complexity of the system and the specific processes it undergoes. Our method leverages practical and readily available formulations, ensuring both feasibility and applicability. The novelty lies in upscaling permeability−porosity evolution while accounting for associated uncertainties and focusing on permeability

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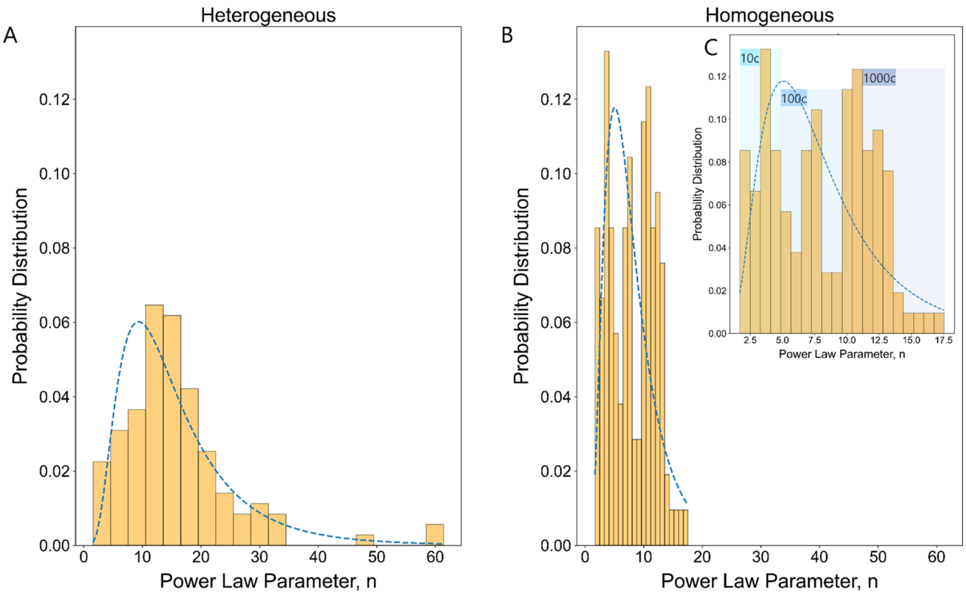

Figure 6. Probability distribution function (PDF) and statistical variance of power-law exponent (n) for: (A) heterogeneous structures; (B) homogeneous structures with the x-axis comparable to A; and (C) homogeneous structures with different crystallization cases (10, 100, and 1000) highlighted.

changes induced by mineral precipitation. Rather than employing a generic clogging model across all simulations, our approach involves developing a tailored porosity−permeability relationship that is fit-for-purpose and specific to the type of pore structure and prevailing physical/chemical conditions. Regarding pore structure type, we categorize lithologies based on shared characteristics such as sorting, grain sizes, and void connectivity, distinguishing between homogeneous clastic and heterogeneous nonclastic 2D lithologies—most pertinent for media like fractures. A comprehensive set of litho-types will be proposed in future research, but for now, we've focused on outlining the general features of our hierarchical upscaling scheme.

The power-law relations have been typically used to describe the effects of mineral precipitation on various processes altering pore morphology such as biomass accumulation, dissolution, and precipitation. As highlighted by Hommel et al.,[7] more intricate modifications to permeability−porosity relationships do not necessarily yield more accurate predictions compared to simple power-law equations with appropriately chosen exponents. This observation stems from the fact that many of these more complex relations either include a power-law term or exhibit similar behavior. Furthermore, the simplicity and versatility of the power-law equation render it well-suited for integration into numerous flow simulations and reactive transport modeling (RTM) tools. Hence, in our approach, we extract the coefficients "n" and "a" by fitting the power-law curve described by eq 24 to the data from each realization, as depicted by individual trendlines in Figure S3.

$$\frac{k}{k_0} = a\left(\frac{\varphi}{\varphi_0}\right)^n, \quad 0 < a \leq 1 \tag{24}$$

The distribution of "n" values is illustrated in Figure 6 and supplemental Figures S8−S11, while the distribution of "a" values is depicted in supplemental Figures S12−S16. In Figure 6, we juxtapose the distributions of "n" values for all the heterogeneous cases (A) and all the homogeneous cases (B,C). For ease of comparison, Figure 6A,B is plotted with identical x,y-axis intervals. The results indicate a notably broader distribution of the power-law parameter in heterogeneous structures, with significant overlap among the heterogeneous cases, making it challenging to distinguish crystallization cases. Conversely, a more structured pattern in the homogeneous cases is evident (Figure 6B,C). The distribution reveals a trimodal distribution, wherein the three peaks could be attributed to different crystallization cases (10, 100, and 1000C), as highlighted in Figure 6C.

In eq 24, the parameter "a" serves to modify the traditional power-law relationship. It is a scaling factor that adjusts the curve to better match observed clogging behavior. When analyzing the distribution results, depicted in Figures S12−S16, it is evident that the parameter "a" tends to cluster around a value of 1 across most cases. This observation implies that the standard power-law formulation with parameter "a" set to 1 adequately captures the behavior of the system. However, a notable exception arises in cases involving 1000 crystallites. In such cases, an intensified initial clogging effect happens, and the mean value of the distribution shifts toward the left side, indicating values of "a" less than 1. This deviation is attributed to a pronounced decrease in permeability resulting from a slight reduction in porosity during the early stages in the 1000 crystallite cases.

The individual results for each case (Figures S8 and S13), structure (Figures S9 and S14), number of crystallites (Figures S10 and S15), and all the collective analysis of entire cases (Figures S11 and S16) offer a comprehensive insight into the clogging behavior observed in the simulated scenarios.

Generating a probability density function (PDF) for the parameters governing the porosity−permeability relationship offers valuable insights into the anticipated clogging behavior, focusing on the mean of the distribution as the most probable behavior. The PDF encapsulates a spectrum from best-case to worst-case scenarios, effectively capturing the intrinsic uncertainty inherent in the porosity−permeability relationship. Derived from pore-scale simulations, this PDF serves as a crucial tool for selecting optimal parameters in clogging models at larger

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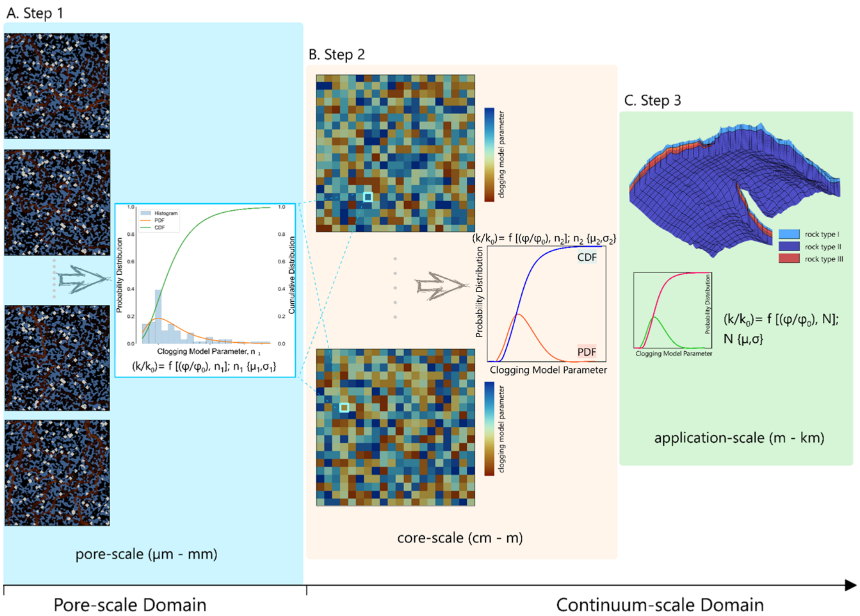

Figure 7. Proposed scheme for upscaling variations in permeability−porosity induced by mineral precipitation and porous media alteration for reactive systems. (A) Pore-scale modeling to capture the clogging behavior of a given system and deriving the probability distribution function (PDF) of the clogging model parameters. (B) Transferring the statistical information to continuum scale. C) Transferring the statistical information to large-scale (application-scale) simulations.

scales. This approach enables explicit acknowledgment and incorporation of the inherent uncertainty associated with the porosity−permeability relationship.

The proposed upscaling framework for incrementally increasing the grid size is illustrated in Figure 7. To enable broader applications in different precipitation processes, the summary here incorporates consideration of the number of secondary crystallites and their spatial distribution as dependent on temperature, supersaturation, and interfacial free energy between the crystal and substrate surface. This results in a randomized distribution of crystallites, as previously detailed in our probabilistic nucleation and growth model.[8,22,39−42] The framework operates as follows:

Step 1: Monte Carlo simulations of crystal growth (refer to Figure 7A)

Pore-scale modeling: The procedure begins with pore-scale modeling, entailing Monte Carlo (MC)-type simulations wherein the positions of a pre-estimated number of crystallites are randomized for each MC iteration. The estimated number of crystallites takes into account factors such as time step, temperature, aqueous supersaturation, and interfacial free energy between substrate and secondary crystals. In the current simplified scheme, the number of crystallites remains constant. However, it is important to emphasize that in more realistic simulations, a robust probabilistic nucleation model[8,40−43,63,64] should determine the number of crystallites and their spatial distribution. In such instances, low interfacial free energies between the secondary crystals and the crystallites (of the same mineral) should give rise to clustering and aggregate formation around pre-existing crystallites, especially at low supersaturations.[40,57] Subsequently, in each MC-simulation crystallites undergo growth based on a specified growth model, such as Burton, Cabrera, and Frank (BCF)[65,66] also depending on the secondary mineral of interest, the aqueous supersaturation and temperature.

Deriving the distribution of clogging model parameters: Porosity−permeability pairs are calculated at various stages during the evolution of the system. Subsequently, by fitting the preferred clogging model curve (in this paper, we utilized the power law) to each simulation, a distribution of the clogging model parameter can be derived. Based on the observations at the pore-scale and the multiplicative nature of the clogging behavior, we anticipate the distribution to follow a (log)normal distribution as more data are accumulated. This step provides a PDF that will be used in the next step.

The data required for the initial step include: A 2D (or 3D) reactive transport pore-scale model, kinetic data for the mineral nucleation−substrate pair, and BCF-kinetic constant for the secondary mineral. While kinetic data are currently lacking for many minerals, efforts will be made to obtain such data for the most crucial minerals of interest.

Step 2: Transfer of statistical information to continuum scale (refer to Figure 7B)

The second step entails a relatively low computational cost as it involves transferring statistical information from the pore scale to grid cells within a continuum model that represents the lithology of interest (the lithology/porous geometry modeled in Step 1). At this scale, each grid cell is assigned a clogging model parameter drawn from the probability distribution in Step 1. Consequently, the majority of grid cells will be populated with parameters near the mean value. The choice of distribution inherited from step 1 is based on the estimated number of crystals formed in the grid cell, which again depends on time step, temperature, and aqueous supersaturation. Longer durations, elevated temperatures, and larger supersaturations all contribute to increased crystal formation.

For a more realistic scenario involving crystal size distribution and cluster formation, clogging model parameters can still be inherited from step 1, provided that the more comprehensive pore-scale simulations were conducted without the simplification of a fixed number of crystallites. Subsequent to populating the grid cells with clogging model parameters, the permeability of each grid cell can be estimated, thereby enabling the calculation of the permeability for the entire system. Iterating this process yields a new probability distribution, tailored to the continuum scale permeabilities. The permeability of the entire domain should be calculated at various points throughout the simulation time frame employing an appropriate averaging technique. The resulting permeability and porosity values will constitute a new PDF for the clogging model parameters.

Step 3: Transfer of statistical information to large-scale (application-scale) simulations (refer to Figure 7C)

The third and final step involves transferring statistical information from the Representative Elementary Volume (REV) scale (step 2) to larger scales comprising different lithologies (pore space characteristics). Each lithology possesses its own upscaled PDF containing the inherited statistical spread of the clogging parameters. these parameters are intricately linked to local (grid-scale) factors such as temperature, aqueous supersaturation, and time, which collectively contribute to estimating the number of crystals.

Although the simplification within the entire framework, where crystals grow uniformly and possess a consistent size at any given time (prior to coalescence) is critical for this model, the outcomes may still hold validity for numerous scenarios. Even in cases where crystals exhibit a nonuniform distribution, this proposed framework represents a significant improvement over the traditional modeling crystal growth at the continuum scale, which typically disregards any information regarding crystal numbers and location, and the corresponding spread of clogging model parameters. Examples of systems where the current scheme may work well is the long-term effect of quartz diagenesis, were syntaxial overgrowth may lead to a uniform clogging of the entire pore space, similar to a case with a very large number of crystals. During CO_2 storage, the formation of local and massive salt aggregates[24,52,67] may on the other hand be a good example represented in the scheme by a few crystals with extensive growth in the porous media. Any system predominantly governed by syntaxial overgrowths may be effectively modeled utilizing the proposed framework.

In majority of RTM purposes, a three-step scheme should adequately capture and transfer the pore-scale probabilistic dynamics to fit-for-purpose application scale. For instance, consider a pore-scale simulation with 500 × 500 grids, akin to the present study, in which the micrometer-scale grid size represents an entire domain of the millimeter scale. Transitioning to the continuum scale (referred to as core-scale in certain applications), the grid size at the millimeter scale encompasses domains ranging from centimeters to meters. Finally, at the application scale, grid sizes of several meters cover the entire kilometer-scale domain (Refer to Figure S13).

By adhering to this upscaling framework, the complexities of permeability changes in porous media can be effectively represented across various scales, while maintaining ease of implementation. Furthermore, the gradual transition from a smaller to a larger scale facilitates the capturing of interactions between consecutive scales, thereby ensuring a more accurate representation of system behaviors.

As we mentioned in our previous publications,[40,42,57] in some specific systems, the probability of nucleation occurring in a particular area and at a specific time is exceptionally high, enabling the employment of deterministic approaches to handle the nucleation process, which significantly simplifies the analysis and modeling.

The subsequent phases will entail upscaling the pore-scale modeling demonstrated in this example to three dimensions (3D) and incorporating the complexity of nonuniform crystal size distributions. To achieve the latter, the initial step involves conducting pore-scale reactive transport simulations that integrate a comprehensive nucleation−growth model, as outlined in.[8,39,42] This process may necessitate iterative runs until a robust statistical sample has been obtained. Consequently, the clogging model parameters will not only be influenced by the number of crystals but also by the distribution of crystal sizes. This additional information must also be transferred to the continuum scale. To streamline this process, a comprehensive database comprising predefined crystal size distributions and corresponding clogging model parameters can be generated by pore-scale RTM. This database serves as a resource for the less computationally intensive steps 2 and 3 within the upscaling scheme.

It is imperative to evaluate the CPU time of the method due to the computational demands associated with performing Monte Carlo-type stochastic realizations. While the incorporation of these larger structural scales falls outside the scope of this study, it represents a focal point for future research endeavors.

5. CONCLUSIONS

In this study, we took a statistical approach to investigate the effect of crystallite distribution on the coevolution of permeability−porosity due to mineral precipitation. We also introduced an upscaling approach to better represent permeability changes at large-scale simulations with implications for clean water and sanitation, climate action, and affordable and clean energy. The following points summarize the major findings of our work:

The spatial distribution of the mineral precipitates and the initial state of the porous geometry play critical roles in changing the hydrodynamics of porous media. Therefore, even with the same process and the same porous morphology, identical porosity changes during mineral precipitation might cause different changes in permeability. Capturing the permeability−porosity evolution

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with a single clogging model might result in considerable uncertainties.

Anisotropic and heterogeneous reservoir rocks may be prone to more severe permeability impairment, as minor changes in porosity can lead to severe permeability impairments.

Fitting the power-law function to each realization resulted in a wide range of power-law coefficients. For most realizations, a (log)normal distribution is obtained. The higher the number of crystallites, the permeability–porosity data are less scattered.

Distribution of the deposited crystallites (controlled by mineral nucleation) adds an extra level of complexity to the mineral precipitation process, which should be taken into consideration in continuum-scale reactive transport simulations. To address this issue, we proposed a hierarchical upscaling approach, by sampling the porosity–permeability power-law coefficients from the probabilistic distributions derived at the finer scale.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.energyfuels.4c01432.

Initial transport properties of the studied structures; list of routinely used porosity–permeability relationships; schematic of the subgrid implementation for crystal growth; model validation; porosity–permeability evolution curves; sensitivity analysis of reaction rate; visualization of velocity field and crystal formation; statistical distribution of clogging model parameters as probability distribution functions; and schematic of microscopic dynamics communication with macroscopic application domain for constructing fit-for-purpose clogging model (PDF)

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Notes

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